



Assignment - 4: Access Control Level (ACLs)

Objective:

The objective of this assignment is to design and implement a structured access-control model for Project Management, Resource Assignment, and Skill Management within a ServiceNow environment. The goal is to ensure that users can interact only with the data relevant to their roles, while maintaining strict security around sensitive records. This includes defining record-level and field-level permissions, enforcing visibility rules, and applying the “least privilege” principle so that every user—Managers, Developers, QAs, Architects, and Operations—has only the access required to perform their responsibilities effectively. This assignment also helps establish consistency, data integrity, and governance across all project-related and skill-related modules in the system.

Pre-requisite – Role Creation:

A new role Resource Manager must be created.

This role is responsible for managing resource assignments and skill mappings without having full project or system control.

1. Access – Managers can edit only their own projects

Managers should have the ability to update project records only when they are assigned as the Project Manager for that specific project.

They must not be able to modify any project that is managed by another Manager or Operations member.

They should not be able to view projects they are not managing..

2. Access – Managers can create and read assignments only for their own projects

Managers should have the ability to:

- Create resource assignments—but only for projects where they are the designated Project Manager.
- Read assignments that belong to the projects they manage.
- Assignment creation is permitted only when the Project Status is Planned or In Progress
- They must not be able to create or read assignments related to projects managed by others.

3. Access – Resource users see only their own project assignments

Team members such as Developers, QAs, and Architects should only be able to view the resource assignments in which they are assigned.

They should not have visibility into:

- Assignments belonging to other team members
- Assignments for projects they are not part of

Their access to assignments is strictly self-based visibility.

4. Access – Only Operations can manage Skills and Skill Mappings The Skills table and the Skill Mapping table should be fully controlled by the Operations team only.

Operations users have full CRUD rights:

- Create skills
- Update skill details
- Delete skills
- Manage employee skill mappings

All other roles (Managers, Developers, QAs, Architects) must have no access (no create, read, update, or delete).

5. Access – Users see only their own Skill Mappings

Employees should only be able to view their own skill profiles.

For example:

- A Developer can see only skills mapped to themselves
 - They cannot view skill mappings of other Developers, QAs, Architects, or Managers
- Operations, however, can view and manage all mappings.

6. Field Security – Unique ID fields must be read-only for all non-Operations users Project ID, Assignment ID, Skill ID, and Skill Mapping ID should be read-only to ensure no unauthorized modification.

7. Field Security – Project Status should be editable only by Operations

Only the Operations team should have permission to update the “Project Status” field. Managers should not be able to directly change the status.

8. Access – Resource Manager can create Resource Allocations

Resource Managers should be able to create new Resource Allocation records.

- Can read all project records
- Cannot update or delete projects
- Cannot delete any record in the system

This role acts between Operations and Managers, focusing only on staffing activities.

9. Access – Resource Manager can create Employee Skill Mappings

Resource Managers should be able to create new Employee Skill Mapping records.

- Can create employee skill mappings
- Can read skill and employee records
- Cannot delete skill mappings

Skill governance remains controlled while allowing onboarding of skills.

10. Project – Required Resources Fields Locking

The Proficiency Level field in the Employee Skill Mapping table must be strictly controlled.

Fields:

- Developers Needed
- QA Needed
- Architects Needed

Requirement:

1. Editable only when Project Status = Planned
2. Once status ≠ Planned, fields become read-only
3. Operations can edit these fields at all times

11. Access – Only Operations can edit Proficiency Level

The Proficiency Level field in the Employee Skill Mapping table must be strictly controlled.

- Only Operations can update proficiency level
- All other roles have read-only access
- No other role can modify this field

12. Access – Assignment Update Restriction After Approval

Once an Assignment reaches a final state, further modifications must be restricted.

- Assignments with Status Approved or Rejected must not be updatable
- Only the Operations role may update such assignments
- All other roles have read-only access after final status is set